Testing and Signal Generation Methods

With so many failures in the development process, it is crucial that failing ideas be weeded out as early as possible, freeing up time for potential winners. The preliminary steps towards success are:

Establish a testing regimen: There are four basic ways to generate trading signals: historical backtesting, out-of-sample, walk forward, and real-time analysis.

1. **Historical backtesting** is the most common approach, and traders love it since it generally produces the best results — smooth equity curves, modest drawdowns, strongest risk-to-reward metrics. And with a little optimization, the results look even better — so much better that they cannot be achieved in real life. Gently optimizing does have a major advantage in eliminating ideas early on. Results will never be better, so use this testing approach to screen out ideas early in the process.
2. **Out-of-sample testing** helps solve the optimization problem by breaking your data into two pieces: in-sample and out-of-sample. Assume you have five years of data, and that you gently optimize the first three years of in-sample data and then use the optimized parameters for the two years of out-of-sample data.
3. **Walk forward analysis** (WFA) is a backtesting technique that involves optimizing a trading strategy on a segment of historical data (the in-sample period), then testing it on a subsequent segment of unseen data (the out-of-sample period). This process is repeated multiple times to simulate real-time trading and to check if the strategy can adapt to changing market conditions.
    a. **More realistic outcome:** WFA produces a more realistic performance assessment than the standard optimized backtesting by more closely simulating real trading conditions. While intraday models often see similar results comparing WFA with optimized backtests, swing models can be quite different, since positions are held days, weeks, or even months.
    b. **Adapts to changing market:** Unlike optimized backtests, which are static by their very nature, WFA adapts to changing market conditions as it rolls forward. This can be seen in Figure 12.1.1 The yellow curve is SPY. The vertical lines are the out-of-sample periods. The blue curve is the walk forward analysis, which was generally

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adapted quickly to the market, maintaining a positive return with minimal drawdowns.

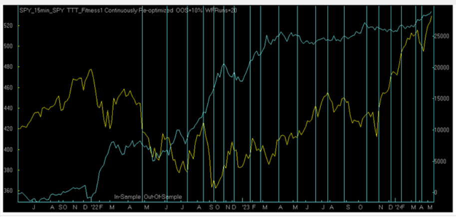

FIGURE 12.1.1 WFA example: yellow curve marks SPY, blue curve marks WFA, blue vertical lines mark out-of-sample periods

i. **Identify effective strategies:** Not only can WFA help identify strategies that are overly fit, but they can also identify those that are generally effective.

ii. **Go/no-go:** This is another decision point; if the WFA does not produce positive results that meet your objectives, discard it and move on.

4. **Real-time analysis** is the most accurate, but it has a major drawback — it can take weeks, months, or even years to have enough trades to produce a valid analysis. Statisticians cannot seem to agree on how many trades you need to generate meaningful statistics. 30-50 seems to be a consensus, but more is always better.

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Monte Carlo

This is the last step in analyzing the trading system before establishing position size to meet risk targets. Monte Carlo analysis is a powerful statistical method used in trading to model the probability of different outcomes in a process that is inherently uncertain, such as the returns of a trading strategy. By simulating many possible future price paths or portfolio outcomes, traders can get a better understanding of the potential risks and rewards involved. Here is how it works in the context of trading:

1. **Purpose of Monte Carlo in trading**
   - **Risk management:** Monte Carlo analysis is often used to estimate the range of potential losses or profits under different market conditions. This helps traders manage risk and optimize their trading strategies.
   - **Probabilistic forecasting:** It allows traders to forecast returns based on statistical distributions, instead of relying on a single point estimate.
   - **Strategy testing:** Traders can simulate the performance of trading strategies under various market scenarios and see how the strategy performs over time.
2. **How Monte Carlo simulations work**
   - **Step 1: Define the model**
     The first step is defining a mathematical model for the trading system that reflects how its price or return might behave. The model often includes assumptions about expected returns, volatility, and correlations with other assets.
   - **Step 2: Random sampling**
     In Monte Carlo analysis, the key is to simulate many different scenarios by randomly sampling from the assumed distributions. These could be normal distributions for asset returns, log-normal distributions for price movements, etc.
   - **Step 3: Run the simulations**
     Thousands or even millions of different potential future price paths are generated based on the random sampling. Each path represents a possible future outcome. Over time, patterns emerge, showing likely and extreme outcomes.
   - **Step 4: Analyze the results**

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- Worst-case losses
- Probability of reaching specific targets or thresholds
- Impact of various levels of volatility

3. **Applications of Monte Carlo in trading**
   Trading strategy evaluation: Before deploying a strategy in the market, traders can run Monte Carlo simulations to understand how the strategy might perform in different market scenarios and across various time frames.

4. **Key assumptions**
   - **Randomness:** Monte Carlo assumes that the future price movements are random and based on probabilistic rules, often following a normal or log-normal distribution.
   - **Stationarity:** Many models assume that the statistical properties of the asset (e.g., volatility and mean return) remain constant over time, although more sophisticated models can account for changing market conditions.

5. **Limitations**
   - **Model dependency:** Monte Carlo results are only as good as the model assumptions. Poor assumptions about asset return distributions, correlations, or volatility can lead to misleading results.
   - **Computational complexity:** Running large-scale simulations can require significant computational power, especially for complex portfolios or derivative structures.
   - **Path dependency:** In certain financial instruments (e.g., path-dependent options), the value depends on the sequence of price movements, adding complexity to the simulation.

Going Live

Once you are satisfied with the performance of your trading system, you can start paper trading live. Monitor the markets closely, adhere to your trading plan, and be prepared to make adjustments as needed based on evolving market dynamics.

1. **Tiptoe in with real money:** When you have at least 50 live simulated trades, go live in a real account with the minimum position size possible.
2. **Increase trade size:** Increase your position size if your returns are as expected after each 30 trades until you reach the maximum position size.
3. **Evaluate and refine:** Continuously monitor the performance of your trading system and make adjustments as necessary. Keep detailed records of your trades and analyze the results to identify areas for improvement.

Remember that developing a successful trading system takes time, patience, and dedication. It is important to approach

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Strategy Profiles

Calculating the expectancy for different trading strategies involves understanding the unique characteristics and risk profiles of each strategy. Different trading strategies have unique profiles that affect their expectancy. Traders must understand these profiles to manage risk and optimize their strategies. Regularly calculating and reviewing expectancy helps traders make informed decisions and adapt to changing market conditions.

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The Seven Steps to a Consistently Profitable Trading System

1. **Trading objectives** should be documented up front — it is much more difficult to accept a marginal system if it doesn't meet all our objectives.
2. **Describe the trading idea**, including a general description, markets to test, bar intervals, historical test period, entry and exit rules, and risk management.
3. **Basic testing** relies on the data from the trading idea. This phase uses lightly optimized testing and exhaustive or genetic testing depending on the number of variables.
4. **Walk forward testing** is the core testing approach since it generally adapts to changes in the market, resulting in better results. If this phase does not produce results matching your objectives, discard it and start over, since it will only get worse from here.
5. **Monte Carlo analysis** is a very difficult test, and otherwise good systems can and do fail at this point, either because returns come under pressure or drawdowns expand.
6. **Live trading** in a simulated account or very small positions in a real account are the acid test, and they must meet your objectives. If they do, increase position size a little at a time until you reach your maximum position size.
7. **Add leverage** — or deleverage — based on the maximum drawdown data.

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